MILITARY SPECIFICATION

CATECHOL, 4-TERTIARY BUTYL

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification covers one grade of 4-tertiary butyl catechol commercially known as the practical grade.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Federal

FED. STD. NO. 141	Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application

FSC 6810

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to the Commanding Officer, Naval Supply Depot (Code CDI), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 No data is required by this specification or by applicable documents referenced in Section 2, unless specified in the contract or order.

3.2 Material - The 4-tertiary butyl catechol shall be a product of high quality, suitable for the purpose intended, and so manufactured as to meet the requirements specified herein.

3.3 Physical properties - The 4-tertiary butyl catechol shall conform to the physical properties listed in Table I.

TABLE I - PHYSICAL PROPERTIES

CHARACTERISTIC	REQUIREMENT	TEST PARAGRAPH
Color	White to light brown	4.4.1
Freezing point, min.	50.0°C	4.4.2
Boiling point, min.	280°C	4.4.3
Pounds per gallon, 60°C	8.68-8.78	4.4.4
Flash point, C.O.C., min.	300°F	4.4.5
Fire point, C.O.C., min.	330°F	4.4.5

3.4 Workmanship - The material shall be uniform in quality and shall conform to the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Acceptance inspection - Conformance of the material to the requirements of this specification shall be determined entirely by means of acceptance inspection. The acceptance inspection shall consist of an examination for acceptability of quality control methods used by the manufacturer, examining and testing the acceptance samples (4.3.2) for all of the requirements of this specification, and an examination of the sample of filled containers (4.4.3) for conformance to the packaging, packing and marking requirements.

4.3 Sampling -

4.3.1 Size of lot - For the purpose of sampling, a lot of 4-tertiary butyl catechol shall consist of a manufacturer's batch. If the material cannot be identified by batch, a lot shall consist of not more than 1,000 pounds of 4-tertiary butyl catechol offered for delivery at one time.

4.3.2 Sample for tests - From each lot offered for acceptance under contract, two 1 pound samples of 4-tertiary butyl catechol shall be removed from separate unit containers taken at random. If more than one batch is included in the lot, two random 1-pound samples shall be taken from each batch.

4.3.3 Sample for examination of filled containers - A random sample of filled containers shall be selected from each lot of 4 tertiary butyl catechol offered for acceptance under contract, in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) = 2.5 percent defective.

4.4 Inspection methods -

4.4.1 Visual inspection - Conformance of the 4 tertiary butyl catechol to the requirements for material (3.2), color (3.3) and workmanship (3.4) shall be determined by visual inspection.

4.4.2 Freezing point - The freezing point shall be determined using a 22 by 175 mm.. test tube and 75 mm. immersion thermometer. Place a sufficient amount of the sample in the test tube to fill it to a depth of 100 mm. when molten. Stir continuously with the thermometer, allowing the sample to slowly cool to 48°C. When the temperature of the molten sample reaches 48°C., seed with crystals of 4-tertiary butyl catechol having a known freezing point of minimum 50°C. Continue stirring as the material crystallizes. The highest temperature is recorded as the freezing point.

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4.4.3 Boiling point - The boiling point shall be determined using any suitable boiling point apparatus.

4.4.4 Pound per gallon - Pounds per gallon shall be determined by Method 4184 of FED-STD-141.

4.4.5 Flash and fire points - Flash and fire points shall be determined by Method 4294 of FED-STD-141.

4.5 Rejection criteria - If a sample fails to meet any of the test requirements of this specification, the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging -

5.1.1 Level A - Not applicable.

5.1.2 Level B - Not applicable.

5.1.3 Level C -

5.1.3.1 Cleaning, drying and preservation application - Not applicable.

5.1.3.2 Unit packaging - Unless otherwise specified, unit packaging of the material shall be in clean, dry containers of tne size and type specified by the purchase order.

5.2 Packing -

5.2.1 Level A - Not applicable.

5.2.2 Level B - Not applicable.

5.2.3 Level C - Material packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking -

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5.3.1 Special markings - Each container shall have the following special marking:

WARNING
HAZARDOUS SOLID

Avoid contact with skin, eyes and clothing.

Use goggles, or face shield and protective clothing.

In case of contact, immediately flush skin or eyes with plenty of water for at least 15 minutes; get medieal attention.

Remove and wash clothing before reuse.

5.3.2 Normal markings - In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use - 4-tertiary butyl catechol covered by this specification is intended for use in explosives.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantity required in pounds.
(c) Applicable levels of packing and marking with requirements in detail, if other than as specified in Section 5.
(d) Size and type of container desired.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-81350(WP) CATECHOL, 4-TERTIARY BUTYL

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0-6878

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Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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Naval Air Engineering Center
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